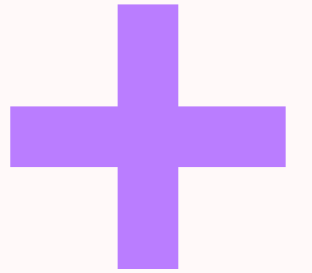
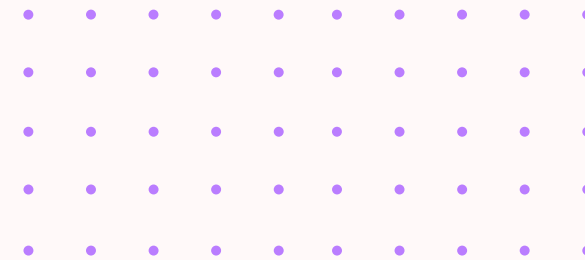
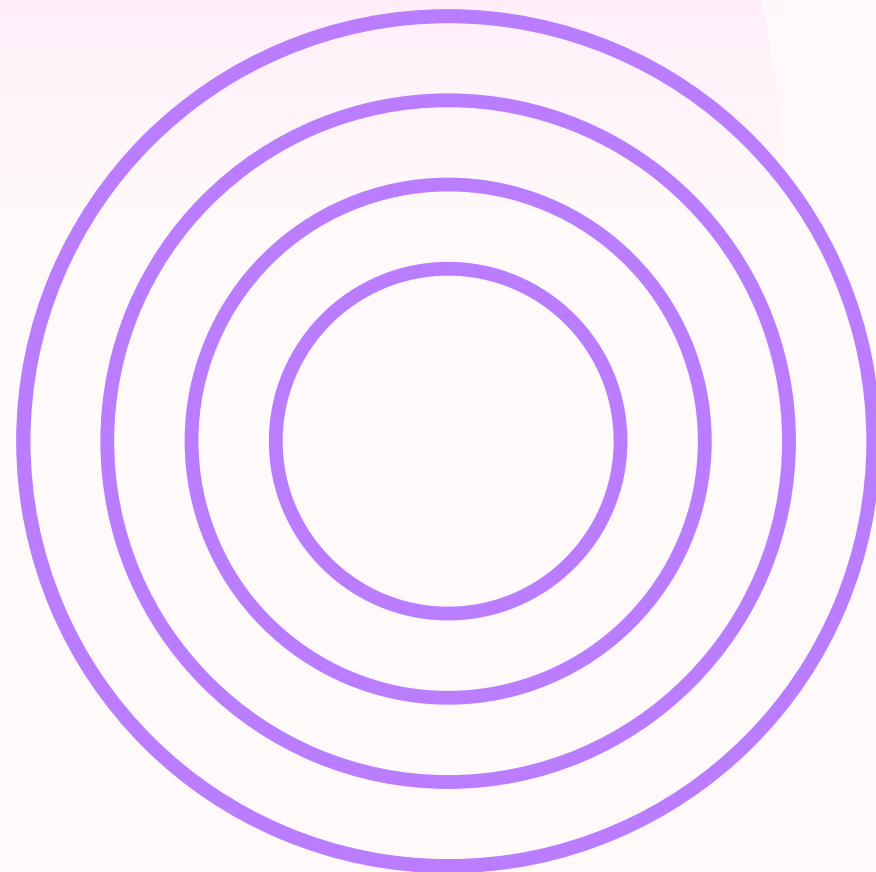


Healthcare Monitoring Device Network



Team Members:
B.Tarun
K.Tejomai



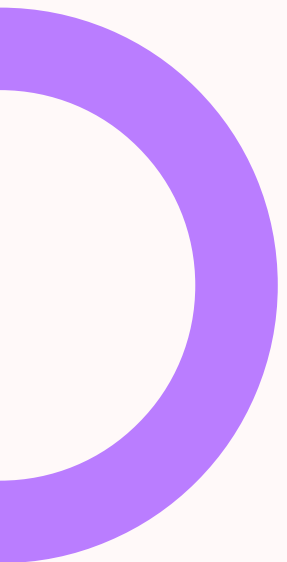
DSA PROJECT

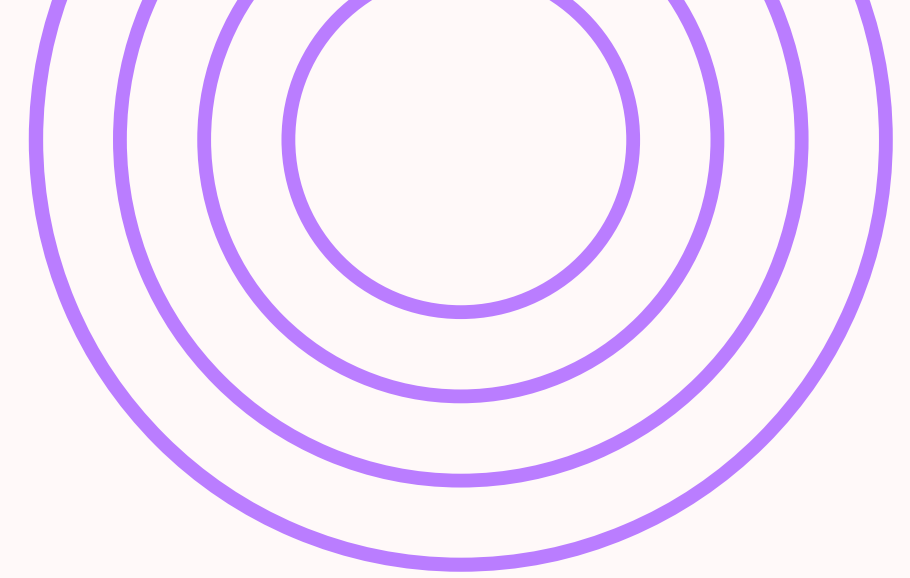
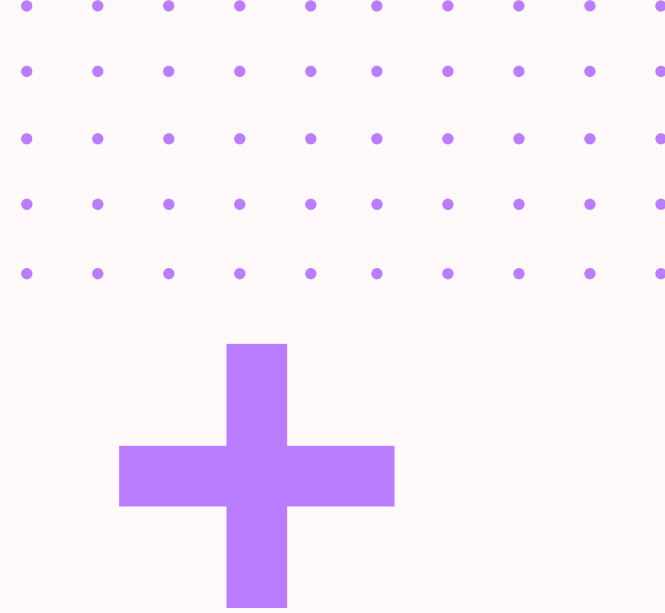


Overview:

This project implements a Healthcare Monitoring Device Network using the Graph Data Structure in C. It simulates how medical devices (like heart monitors, oxygen sensors, etc.) are connected and managed within a hospital network.

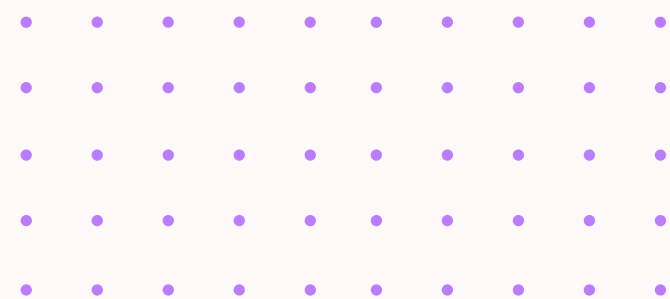
The system supports full CRUD operations (Create, Read, Update, Delete) on devices and allows visualization of their network connections using an adjacency matrix.

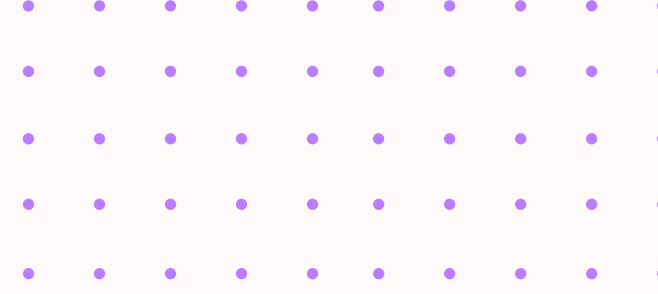




Features:

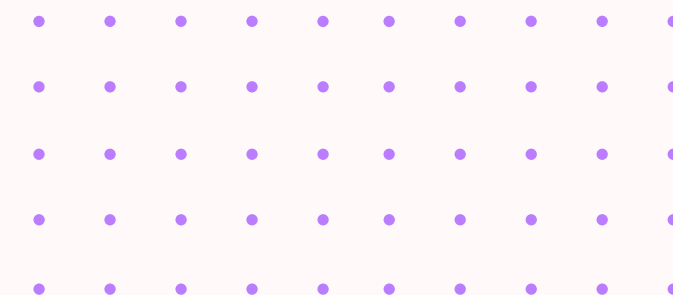
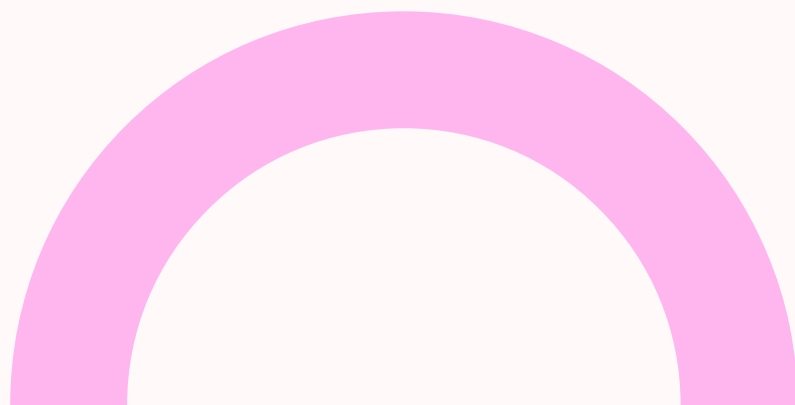
- Add new medical devices (Create)
- View all registered devices (Read)
- Update device details (Update)
- Delete devices from the network (Delete)
- Create connections between devices
- Display network graph (Adjacency Matrix)





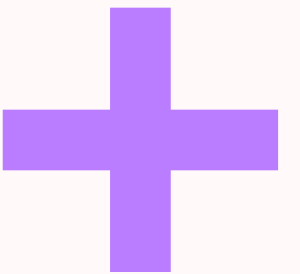
Data Structures Used:

- Structure (struct) → To store device details
- Graph (Adjacency Matrix) → To represent connections
- Arrays → For storing devices and network relationships



Technologies:

- Language: C
- Compiler: GCC / Turbo C / Any standard C compiler
- Concept: Graph Theory + CRUD Operations





How It Works:

Each device contains:

- ID → Unique identifier
- Name → Device type (e.g., ECG, BP Monitor)
- Status → Active / Inactive / Fault
-

Connections between devices are stored in an adjacency matrix, where:

- 1 = Connected
- 0 = Not connected



How to Run:

1. Copy the code into a file:

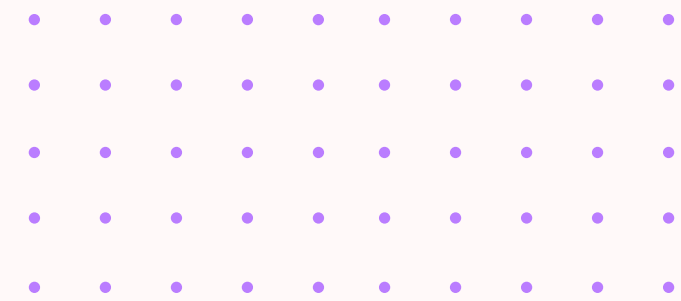
healthcare_network.c

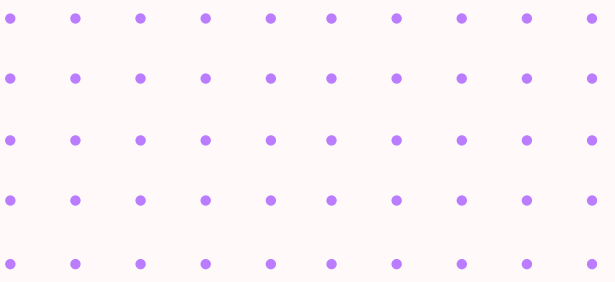
2. Compile the program:

gcc healthcare_network.c -o network

3. Run the executable:

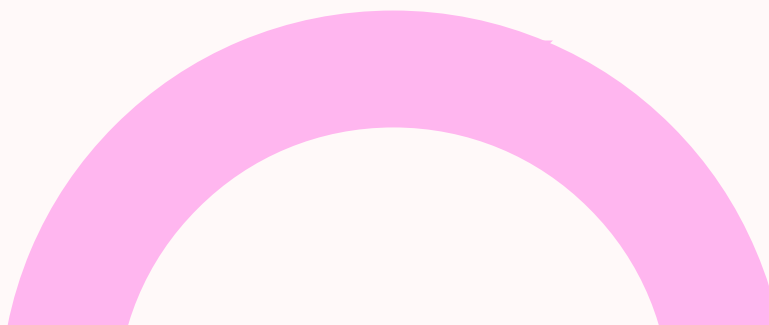
./network

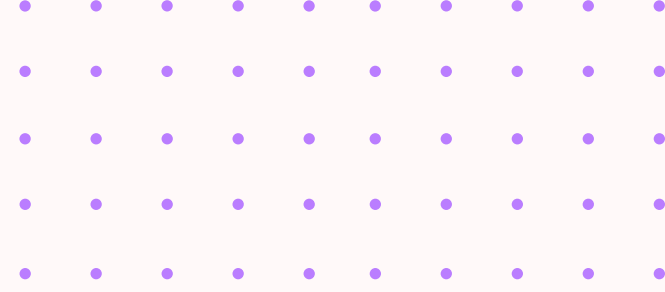
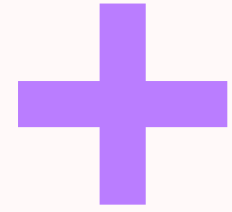




Sample Menu:

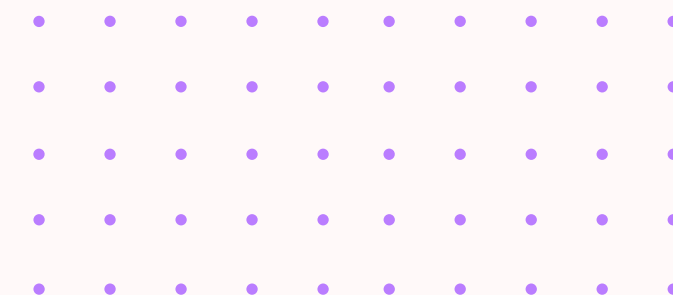
--- Healthcare Device Network ---

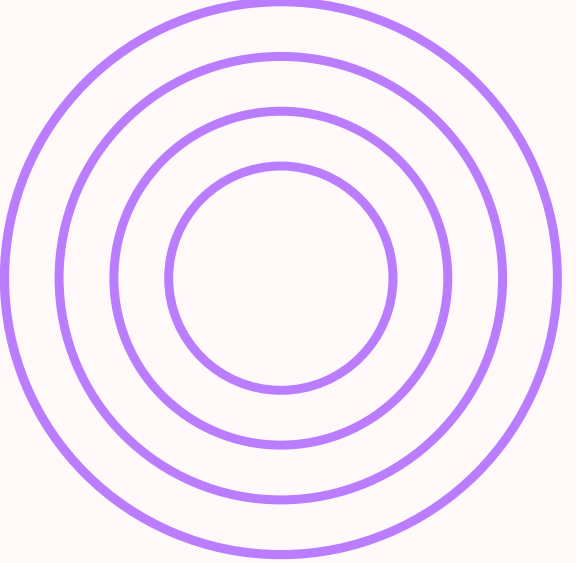
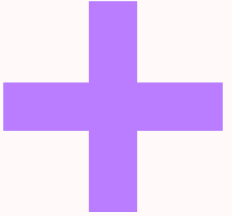
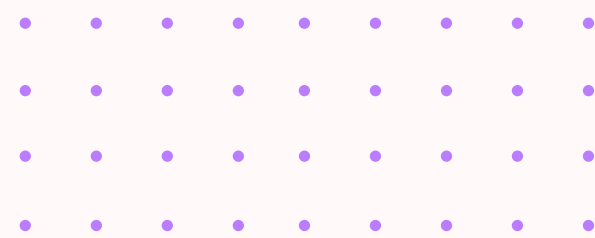
- 
1. Add Device
 2. View Devices
 3. Update Device
 4. Delete Device
 5. Add Connection
 6. View Network
 7. Exit
- 



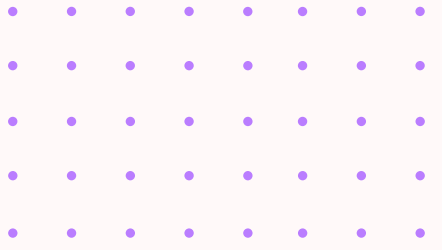
Example Use Case:

- Add devices like:
- Heart Monitor
- Oxygen Sensor
- Temperature Sensor
- Connect them to simulate a hospital ICU network
- Monitor device status and connections





Future Enhancements:

- Implement BFS/DFS traversal
 - Add file storage (persistent data)
 - Real-time alert system
 - Convert into GUI application
 - Database integration
- 

*THANK
YOU*

